Automotive Transportation Operations

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1. Executive Summary

An assessment of **11,200 transportation orders**, including **10,000 completed deliveries**, identified significant operational performance gaps.

The on-time delivery rate was **65.94%**, compared with an average target of **92.18%**, resulting in a **26.24-percentage-point performance gap**.

The assessment also identified prolonged carrier assignment response times, variations in carrier and route performance, and **254 active overdue shipments**.

Management Priority: Improve carrier assignment efficiency, delivery reliability, and proactive exception management.

2. Key Performance Indicators

KPI	Result
Total Transportation Orders	11,200
Completed Deliveries	10,000
On-Time Delivery Rate	65.94%
Late Delivery Rate	34.06%
Average On-Time Target	92.18%
Performance Gap	−26.24 pp
Average Delivery Duration	47.81 hours
Overdue Active Shipments	254
Rejected Assignment Attempts	1,277
Average Accepted Assignment Response Time	590.51 minutes

3. Key Operational Findings

Carrier Performance: Horizon Car Haulers recorded the highest on-time delivery rate at 69.00%, compared with 63.17% for Desert Fleet Logistics. All six carriers performed below their respective service targets.

Routes & Congestion: High-congestion shipments recorded a 40.43% late delivery rate, compared with 30.88% under low congestion. The Riyadh–Jazan route recorded a 48.98% late delivery rate across 49 completed deliveries.

Assignment Efficiency: The average accepted assignment response time was 590.51 minutes. Only 5 of 11,200 accepted assignments met their recorded response-time targets, indicating the need to review assignment procedures and SLA definitions.

Operational Exceptions: The assessment identified 254 overdue active shipments and 59 assignment records with negative response durations requiring validation.

These findings identify operational risks and areas for investigation but do not independently establish the causes of delivery delays.

4. Proposed Process Improvements

The current transportation process (**AS-IS**) was evaluated against a proposed improved workflow (**TO-BE**).

Operational Area	Proposed Improvement
Carrier Assignment	Monitor SLA compliance and establish reassignment procedures
Vehicle Dispatch	Identify dispatch delays and trigger operational alerts
Shipment Monitoring	Track shipment milestones and promised delivery deadlines
Overdue Deliveries	Prioritize exceptions and establish escalation procedures
Order Closure	Verify proof of delivery before closing orders

5. Priority Recommendations

1. **Strengthen Carrier Management:** Monitor carrier performance against service targets and comparable route conditions.
2. **Improve Assignment Efficiency:** Standardize response-time expectations, rejection handling, and reassignment procedures.
3. **Implement Proactive Exception Management:** Prioritize overdue shipments and escalate critical cases.
4. **Improve Operational Monitoring:** Review delayed dispatches and high-risk transportation routes.
5. **Enhance Data Quality:** Resolve invalid timestamps and standardize delivery closure requirements.

6. Management Conclusion

The assessment highlights substantial opportunities to improve transportation reliability and operational coordination.

Final Recommendation: Prioritize carrier response controls and overdue shipment management, followed by standardized performance monitoring and exception-handling procedures.

The proposed improvements are intended to strengthen operational visibility and support more timely decisions. Their effectiveness should be measured after implementation.

This report is based on synthetic data. The proposed improvements have not been implemented or validated in a live operational environment.